TypoFormer Model Architecture Specification

Md. Fahim Sarker Mridul

October 28, 2026

1 TypoFormer Architecture Diagram (NVIDIA Green Theme)

Below is the complete internal architecture of the **TypoFormer** keystroke companion, featuring dual-channel character-aware token fusion, pre-norm grouped-query attention with rotary embeddings (RoPE), SwiGLU feed-forward networks, and multi-task heads for real-time autocomplete and grammar error correction.

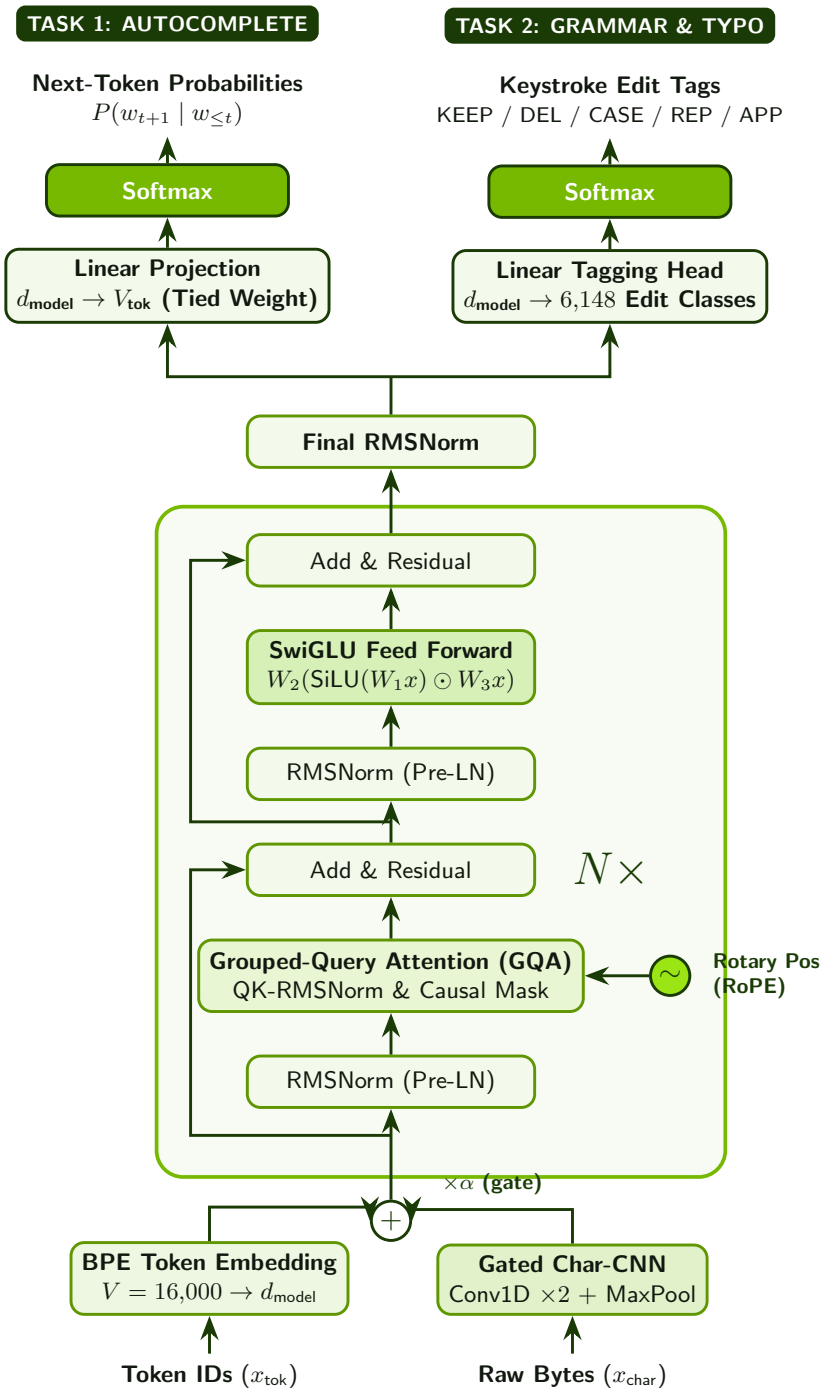


Figure 1: TypoFormer Model Architecture (NVIDIA Green Style): Integrating byte-level subwords with multi-scale character convolutions, pre-norm grouped-query attention, rotary embeddings (RoPE), and joint dual heads for autocomplete and grammar error correction.

2 Classic Transformer Architecture (Vaswani et al., 2017)

For reference and comparison, the exact encoder-decoder architecture from *Attention Is All You Need* is also provided below with identical color grading and geometry.

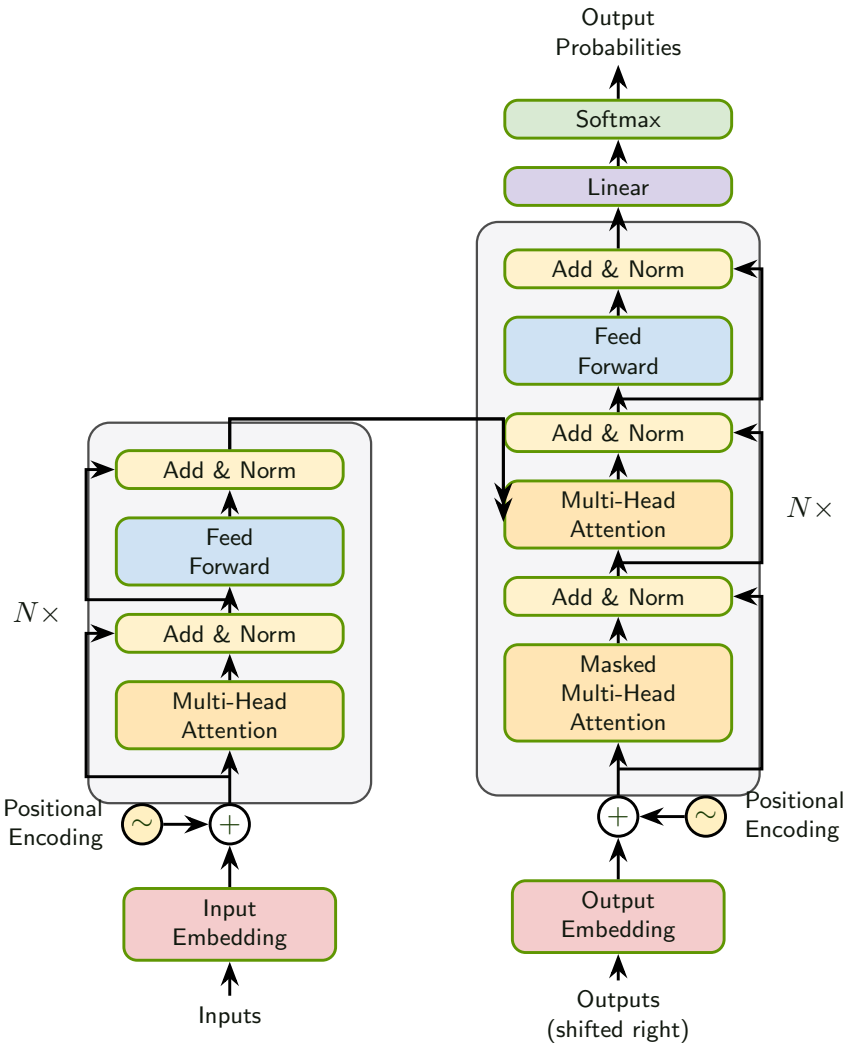


Figure 2: The Transformer – model architecture (Vaswani et al., 2017).